Kafka Core Concepts

Kafka for Microservices Communication

What is Kafka?

Apache Kafka is a distributed event streaming platform.

It is used for:

- Microservices communication
- Real-time data streaming
- Event-driven architecture

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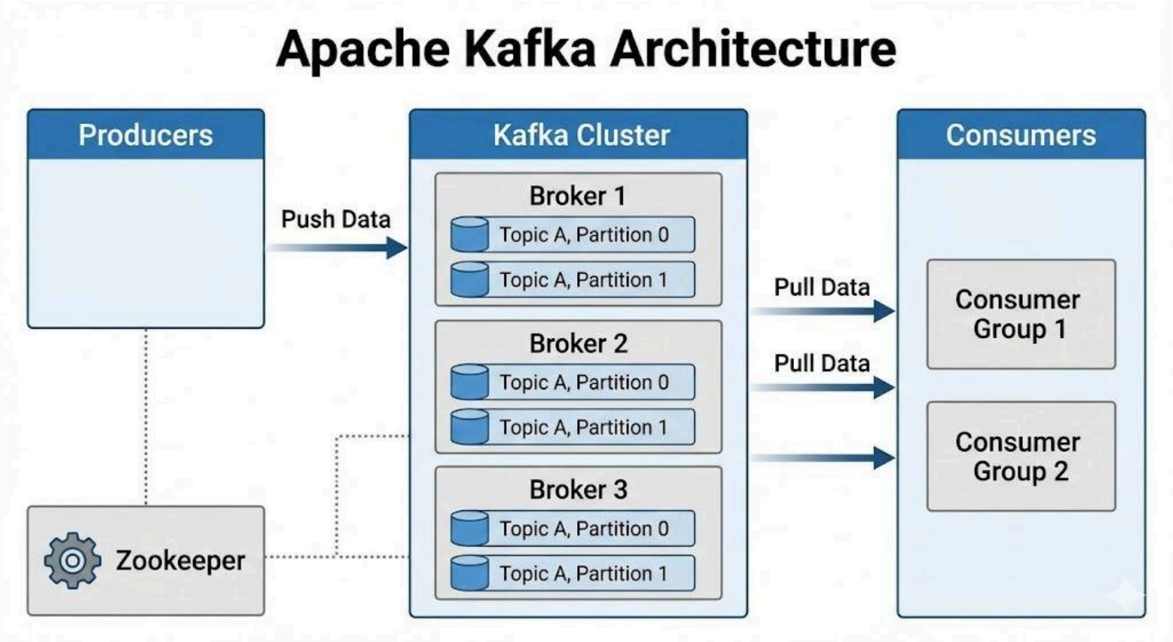
Kafka enables systems to communicate using events instead of direct API calls

Kafka Architecture Overview



Kafka acts as a middle layer between producers and consumers.

Flow:

Producer → Kafka → Consumer

Why Kafka is Needed?

Modern systems require:

- High speed communication
- Scalability
- Fault tolerance
- Loose coupling between services

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Traditional REST calls are not enough for large distributed systems

Producer (Concept)

A Producer is a service that sends data (events) to Kafka.

Example:
Order Service → sends “Order Created” event

Responsibilities:

- Create event
- Send to topic
- Choose partition

Consumer (Concept)

A Consumer is a service that reads data from Kafka.

Example:
Inventory Service → reads order event

Responsibilities:

- Read messages
- Process events
- Update database
- Commit offset

Topic Concept

A Topic is a category where messages are stored.

Examples:

- order-events
- payment-events
- notification-events

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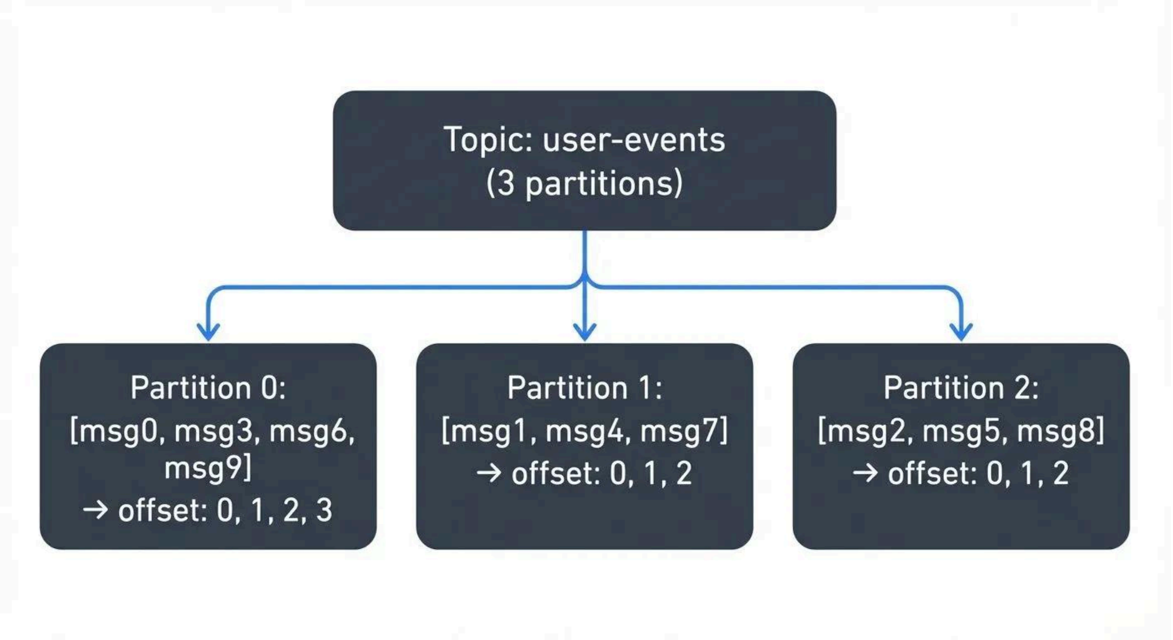
Think of it as a folder for events

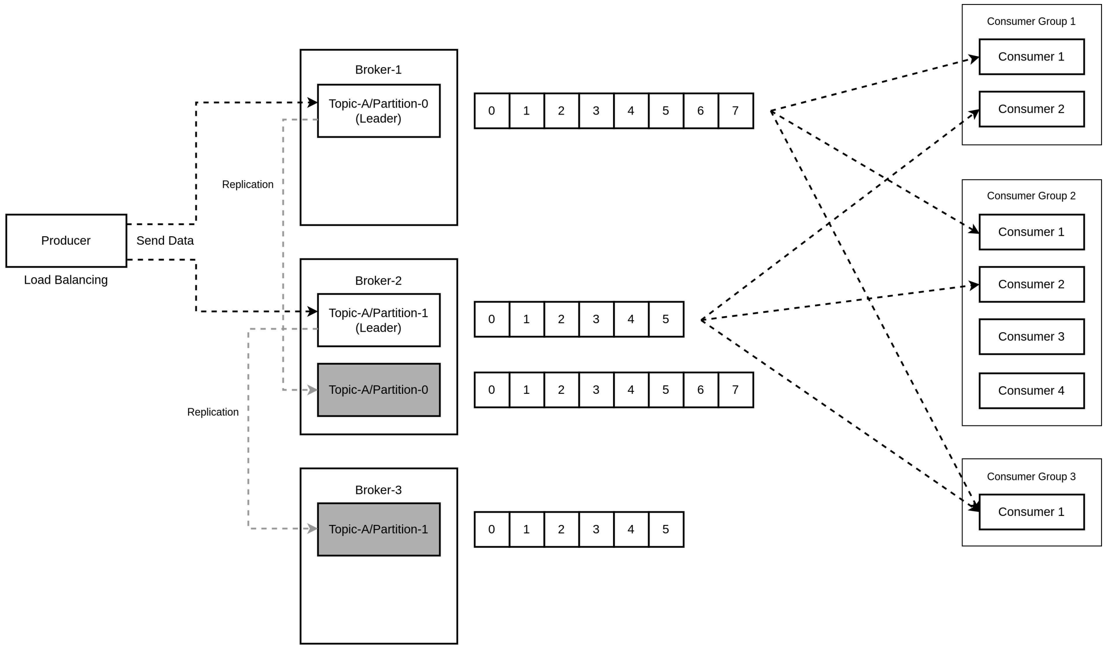
Topic + Partition Model



Topics are divided into partitions for:

- Parallel processing
- Scalability
- Performance improvement

Consumer Group Concept

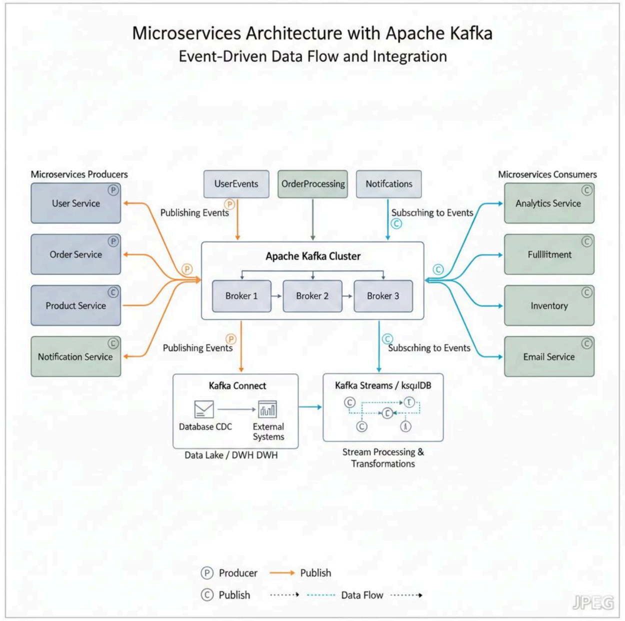


A Consumer Group is a set of consumers working together.

Benefits:

- Load balancing
- Parallel processing

Event Driven Architecture



Services communicate using EVENTS instead of direct API calls.

Event Flow Example

1. Order Service creates order
2. Event is published to Kafka
3. Kafka stores event in topic

Consumers react independently:

- Inventory Service → updates stock
- Payment Service → processes payment
- Notification Service → sends email

Advantages of Kafka

- Loose coupling between services
- High scalability
- Asynchronous processing
- Fault tolerance
- Durable message storage

Real World Use Cases

Kafka is widely used in:

- E-commerce systems
- Banking transactions
- Logging systems
- Real-time analytics
- Notification systems

Key Kafka Concepts Summary

- Producer → sends events
- Consumer → reads events
- Topic → stores messages
- Partition → enables scaling
- Consumer Group → enables parallel processing
- Event-driven → decoupled architecture